TopFT and CohFT

Typeset course notes

1 November 2026

Source. The 22-page handwritten file `topft-cohft-original.pdf`. The order, examples, and moduli-of-curves review of those notes are retained. Source-page references locate each part of the transcription. Notation and necessary hypotheses are made explicit; all diagrams are redrawn in TikZ.

1 Topological field theories

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Let V be a finite-dimensional complex vector space, the *state space*. We write $V^{\otimes 0} = \mathbb{C}$. A connected oriented surface of genus g with n incoming and m outgoing boundary circles has Euler characteristic

$$\chi = 2 - 2g - n - m.$$

A two-dimensional oriented topological field theory (*TopFT*) assigns to it a linear map

$$\omega_{g,n,m} : V^{\otimes n} \longrightarrow V^{\otimes m}, \qquad g, n, m \geq 0.$$

Disjoint unions are assigned tensor products. The following are the connected-surface form of the axioms.

0. Symmetry. The map is invariant under permutations of the inputs and of the outputs: if P_σ and P_τ permute the corresponding tensor factors, then

$$\omega_{g,n,m} \circ P_\sigma = \omega_{g,n,m}, \qquad P_\tau \circ \omega_{g,n,m} = \omega_{g,n,m}.$$

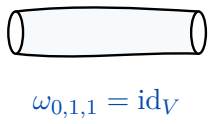
1. Cylinder. $\omega_{0,1,1} = \mathrm{id}_V$.

2. Gluing. Identify $\mathrm{Hom}(V^{\otimes n}, V^{\otimes m}) = (V^*)^{\otimes n} \otimes V^{\otimes m}$. Gluing an output to an input contracts their factors by evaluation $V^* \otimes V \to \mathbb{C}$, or equivalently by the trace on $\mathrm{End}(V)$. There are two cases:

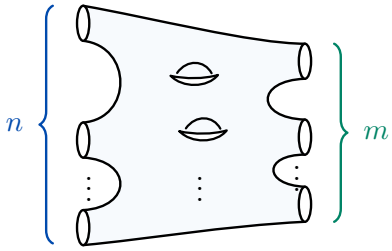
$$\omega_{g,n,m} = \mathrm{tr}_{n+1,m+1}\big(\omega_{g-1,n+1,m+1}\big), \qquad g \geq 1, \tag{1}$$

$$\omega_{g_1+g_2,n_1+n_2,m_1+m_2} = \mathrm{contr}\big(\omega_{g_1,n_1,m_1+1} \otimes \omega_{g_2,n_2+1,m_2}\big)\,. \tag{2}$$

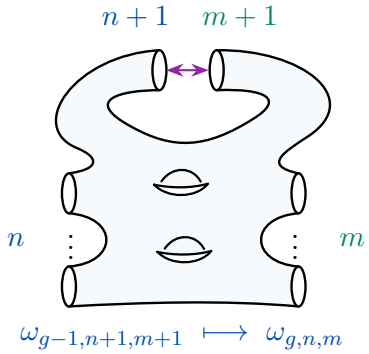
In the second formula the last output of the first factor is glued to the last input of the second. The uncontracted inputs and outputs can be reordered by symmetry.



$\omega_{0,1,1} = \mathrm{id}_V$



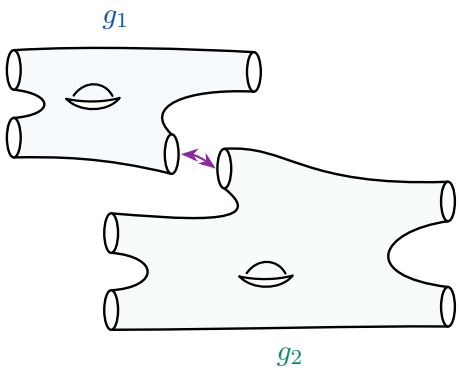
$\omega_{g,n,m}$



Glue the two extra circles.

$\mathrm{End}(V) = V^* \otimes V$
$\xrightarrow{\mathrm{tr}} \mathbb{C}$

A handle is created;
the genus increases by one.



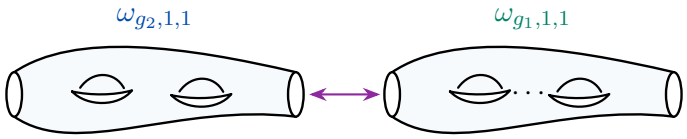
one output↔one input

genus $g_1 + g_2$

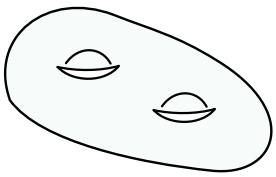
The two pieces become one connected cobordism.

For example, concatenating two surfaces with one input and one output gives

$$\omega_{g_1+g_2,1,1} = \omega_{g_1,1,1} \circ \omega_{g_2,1,1}.$$



Concatenation adds the genera.



"Integration over the space of fields"

V = state space

The theory remembers only
the topology of the surface.

The informal field-theoretic picture is an "integration over the space of fields" on a surface; the topological theory remembers only the topology of that surface.

1.1 Two elementary examples

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(a) A one-dimensional example. Take $V = \mathbb{C}$, so every $\omega_{g,n,m}$ is a scalar. For $c \in \mathbb{C}^*$, set

$$\omega_{g,n,m} = c^{2-2g-n-m}.$$

The cylinder has Euler characteristic zero. If two surfaces are glued along a boundary circle, Euler characteristic is additive, since $\chi(S^1) = 0$. Self-gluing also preserves Euler characteristic. Thus these scalars satisfy the axioms.



Euler characteristic $\chi = 0$

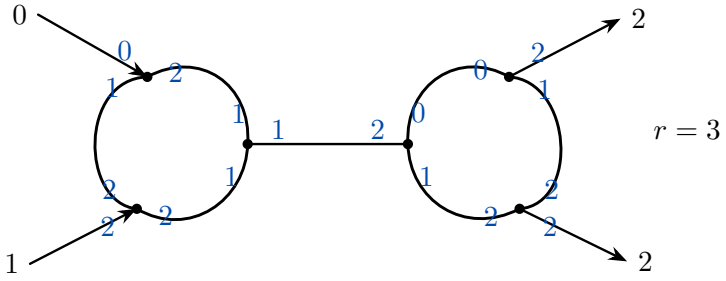
$\chi(S_1 \cup_{S^1} S_2) = \chi(S_1) + \chi(S_2)$

(b) A cyclic-label example. Let $r \geq 1$ and $V = \mathrm{Span}_{\mathbb{C}}\{e_0, \ldots, e_{r-1}\}$, with indices taken modulo r. Define the matrix entries by

$$\left(\omega_{g,n,m}\right)_{a_1,\ldots,a_n}^{b_1,\ldots,b_m} = r^g \, \delta_{\sum_i a_i - \sum_j b_j \equiv 0 \pmod r}.$$

Here δ is 1 when the congruence holds and 0 otherwise, and an empty sum is zero. Thus the sum of the input labels equals the sum of the output labels modulo r.

The notes describe the same rule by labelling a graph obtained from a pants decomposition: the labels at every vertex sum to zero, and the two half-edge labels on an internal edge are opposite modulo r. External input and output labels have opposite sign conventions. Once the external labels satisfy the displayed congruence, there are r^g choices of internal labels, one free choice for each independent cycle.



At each vertex: sum $= 0$. Along each internal edge: sum $= 0$.

Input half-edges carry $-a$; output half-edges carry b.

The displayed labels give one of the 3^2 choices.

This also verifies gluing directly. Self-gluing sums over a repeated label and gives an extra factor r. Separating gluing fixes the glued label uniquely when the total conservation law holds.

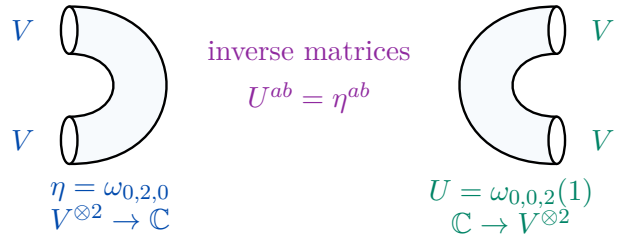
1.2 The pairing, its inverse, and bending boundaries

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Put

$$\eta = \omega_{0,2,0} : V \otimes V \longrightarrow \mathbb{C}, \qquad U = \omega_{0,0,2}(1) \in V \otimes V.$$

The symmetry axiom makes η a symmetric bilinear form.

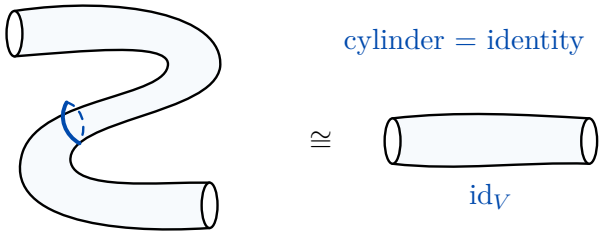


Proposition 1.1. *The maps $\eta^\flat : V \to V^*$ and $U^\sharp : V^* \to V$ determined by η and U are mutually inverse. In particular, η is nondegenerate.*

Proof. Glue a cup and a cap along one boundary. The result is a cylinder, so

$$(\mathrm{id}_V \otimes \eta) \circ (U \otimes \mathrm{id}_V) = \mathrm{id}_V, \qquad (\eta \otimes \mathrm{id}_V) \circ (\mathrm{id}_V \otimes U) = \mathrm{id}_V .$$

For a basis (e_μ) write $\eta_{\mu\nu} = \eta(e_\mu, e_\nu)$ and $U = \sum_{\mu,\nu} U^{\mu\nu} e_\mu \otimes e_\nu$. The identities say $\sum_\nu U^{\mu\nu}\eta_{\nu\lambda} = \delta^\mu_\lambda$. Hence $U^{\mu\nu} = \eta^{\mu\nu}$, the entries of the inverse matrix. □



For the cyclic-label example,

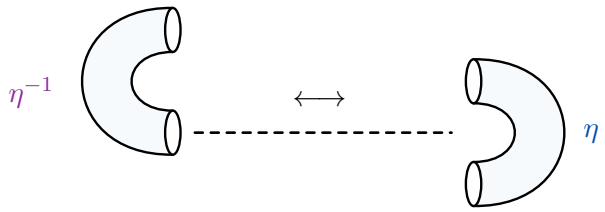
$$\eta_{ab} = \delta_{a+b\equiv 0 \pmod r}, \qquad U = \sum_{a=0}^{r-1} e_a \otimes e_{-a}.$$

Closing the input and output of the cylinder gives the torus. Consequently

$$\omega_{1,0,0} = \mathrm{tr}(\mathrm{id}_V) = \dim V.$$



The pairing identifies an input with an output ("bending a boundary").



Turn inputs into outputs by the inverse pairing,
and outputs into inputs by the pairing.

In particular, all maps are determined by the forms $\omega_{g,k,0}$: for $x \in V^{\otimes n}$ and $y \in V^{\otimes m}$,

$$\eta^{\otimes m}(\omega_{g,n,m}(x), y) = \omega_{g,n+m,0}(x \otimes y).$$

1.3 Finite groups, Hurwitz numbers, and Burnside's formula

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(c) The center of a group algebra. Let G be a finite group and put $V = Z(\mathbb{C}[G])$. If K is a conjugacy class, write $C_K = \sum_{h \in K} h$; these class sums form a basis of V. Define the central element

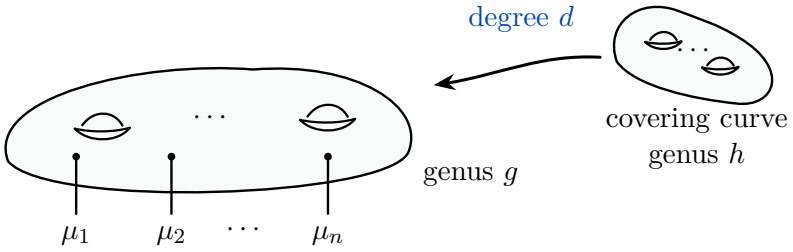
$$\alpha = \sum_{g_1, g_2 \in G} g_1 g_2 g_1^{-1} g_2^{-1}.$$

For $v_1, \dots, v_n \in V$, the theory is specified by

$$\boxed{\omega_{g,n,0}(v_1, \dots, v_n) = \frac{1}{|G|}\,[1_G](v_1 \cdots v_n \alpha^g).}$$

Here $[1_G](z)$ means the coefficient of the identity element in z. The pairing is nondegenerate, so these forms also determine all outputs.

When $G = S_d$, conjugacy classes are indexed by partitions $\mu \vdash d$. The invariant $\omega_{g,n,0}(C_{\mu_1}, \dots, C_{\mu_n})$ is the Hurwitz number of degree-d covers of a fixed genus-g curve, with ramification profiles μ_i over n specified points. The count allows disconnected covering curves and weights each cover by $1/|\operatorname{Aut}|$. Equivalently, it counts tuples of monodromies satisfying the surface-group relation, divided by $d!$; connected covers correspond to the extra condition that the monodromy action is transitive.



Consider degree-d covers ramified over the marked points.
Each μ_i is a partition of d.

$$C_\mu = \sum_{\substack{\sigma \in S_d \\ \text{cycle type } \mu}} \sigma$$

A finite-dimensional commutative $\mathbb{C}$-algebra is *semisimple* if it is isomorphic to a product of copies of $\mathbb{C}$; equivalently, it has no nonzero nilpotent elements. It then has a basis of pairwise orthogonal idempotents, corresponding to $(0,\dots,0,1,0,\dots,0)$. The algebra $Z(\mathbb{C}[G])$ is semisimple, and its primitive idempotents are indexed by irreducible representations ρ of G.

Let $d_\rho = \dim \rho$ and let χ_ρ be its character. On the ρ-summand, a class sum C_K acts by $|K|\chi_\rho(K)/d_\rho$. The resulting *Burnside formula*, for $G = S_d$, is

$$\boxed{\omega_{g,n,0}(C_{\mu_1},\dots,C_{\mu_n}) = \sum_{\rho\in \mathrm{Irr}(S_d)} \left(\frac{d_\rho}{d!}\right)^{2-2g} \prod_{j=1}^{n} \frac{|C_{\mu_j}|\chi_\rho(\mu_j)}{d_\rho}.}$$

Here $|C_\mu|$ denotes the size of the conjugacy class, and $\chi_\rho(h) = \mathrm{tr}(\rho(h))$. The same formula holds for any finite group after replacing $d!$ by $|G|$ and the partitions by conjugacy classes.

1.4 TopFTs and commutative Frobenius algebras

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Definition 1.2. A *commutative Frobenius algebra* is a finite-dimensional commutative associative unital $\mathbb{C}$-algebra $(V,\cdot,\mathbf{1})$ together with a linear functional $\varepsilon : V \to \mathbb{C}$ such that

$$\eta(u,v) = \varepsilon(u\cdot v)$$

is nondegenerate. The functional ε is called the counit.

Not every finite-dimensional commutative algebra admits such a counit. For example,

$$A = \mathbb{C}[x,y]/(x^2,xy,y^2)$$

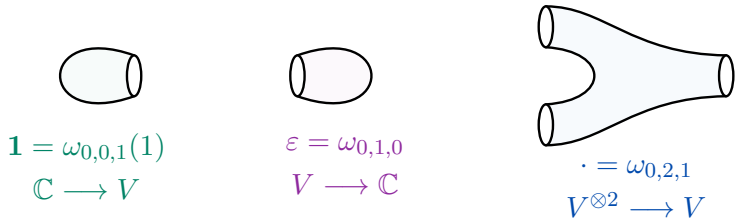
does not. For any ε, its restriction to $\mathrm{Span}\{x,y\}$ has a nonzero kernel vector z. If $a = a_0+a_1x+a_2y$, then $za = a_0z$, so $\varepsilon(za) = 0$ for every $a \in A$. Thus η is degenerate.

Theorem 1.3. *Two-dimensional oriented TopFTs are equivalent to finite-dimensional commutative Frobenius algebras.*

Proof. Given a TopFT, define

$$\mathbf{1} = \omega_{0,0,1}(1), \qquad \varepsilon = \omega_{0,1,0}, \qquad u\cdot v = \omega_{0,2,1}(u\otimes v).$$

These are the disk, the oppositely directed disk, and the pair of pants.

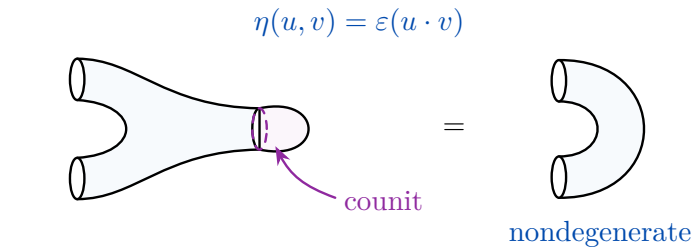


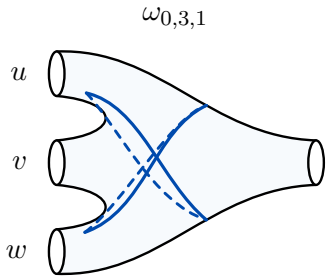
Permuting the two inputs proves commutativity. Gluing a unit disk to a pair of pants gives a cylinder, proving the unit identity. The two decompositions of the genus-zero surface with three inputs and one output give $(uv)w = u(vw)$. Finally, capping the output of the pair of pants shows $\eta(u,v) = \varepsilon(uv)$, and the cup–cap argument proves its nondegeneracy.

Conversely, use multiplication and the unit for the indicated elementary cobordisms, and use ε and the inverse pairing for their oppositely directed versions. Cut a surface into pairs of pants, disks, and cylinders, and compose the corresponding maps. The Frobenius identity

$$\eta(uv, w) = \varepsilon(uvw) = \eta(u, vw)$$

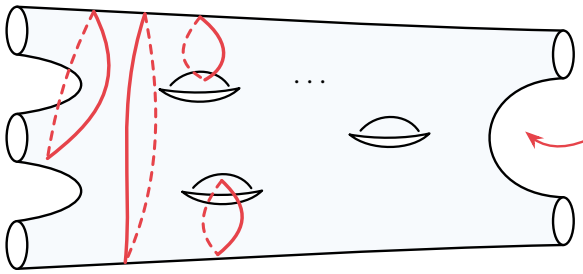
makes multiplication compatible with the pairing. Changing the pants decomposition is generated by the local associativity and Frobenius moves, together with unit and symmetry moves, so the composite is independent of the cut. This produces the TopFT and recovers the original algebra. □

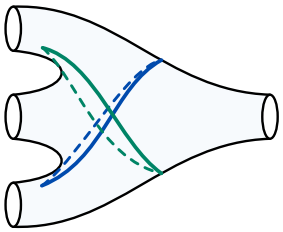




The same four-holed sphere can be cut in two ways.

$(uv)w = u(vw)$





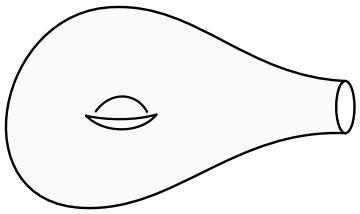
Different pants decompositions give the same $\omega_{0,3,1}$.

$(uv)w$

$u(vw)$

The element that creates a handle is

$$\alpha = \omega_{1,0,1}(1) = \sum_{\mu,\nu} \eta^{\mu\nu} e_\mu \cdot e_\nu.$$



$\alpha = \sum_{\mu,\nu} \eta^{\mu\nu} e_\mu \cdot e_\nu$

by the gluing axiom

$\alpha = \omega_{1,0,1}(1)$

Indeed, glue the two outputs of U to the inputs of a pair of pants. It follows that

$$\boxed{\omega_{g,n,1}(v_1,\dots,v_n) = v_1 \cdots v_n \alpha^g, \qquad \omega_{g,n,0}(v_1,\dots,v_n) = \varepsilon(v_1 \cdots v_n \alpha^g).}$$

For $n = 0$ the empty product is $\mathbf{1}$. These formulas also verify independence of the pants decomposition directly: nondegeneracy and $\eta(u,v) = \varepsilon(uv)$ give

$$\sum_{\mu,\nu} \eta^{\mu\nu} \varepsilon(A e_\mu)\varepsilon(e_\nu B) = \varepsilon(AB), \qquad \sum_{\mu,\nu} \eta^{\mu\nu} \varepsilon(A e_\mu e_\nu) = \varepsilon(A\alpha).$$

The first identity checks separating gluing, and the second checks self-gluing. Raising output indices with η^{-1} recovers the maps with outputs.

The group-algebra pairing and handle element. For $Z(\mathbb{C}[G])$ the counit is $\varepsilon(v) = [1_G](v)/|G|$.

$$C_K = \sum_{h \in K} h \ \in Z(\mathbb{C}[G])$$

sum of all elements
in the conjugacy class K

In the class-sum basis,

$$\eta(C_K, C_L) = \delta_{L,K^{-1}} \frac{|K|}{|G|}, \qquad \alpha = \sum_K \frac{|G|}{|K|} C_K C_{K^{-1}}.$$

For a fixed $b \in K$, its conjugates occur equally often in $\sum_{a \in G} aba^{-1}$, giving

$$\sum_{a,b \in G} aba^{-1}b^{-1} = \sum_K \frac{|G|}{|K|} C_K C_{K^{-1}}$$

a, b range over
all group elements

for fixed b, the conjugates of b
each occur with the same multiplicity

Handwritten aside at this calculation: "Should I do this? (prob. not)"

Thus this is exactly the commutator element used above. For S_d, every conjugacy class is inverse-invariant, so the handwritten formula becomes

$$\eta_{\mu\nu} = \delta_{\mu\nu} \frac{|C_\mu|}{d!}.$$

Recovering Burnside's formula. The idea is to compute the same correlator in the primitive-idempotent basis, where multiplication is coordinatewise. The element p_ρ acts as the identity on the irreducible representation V_ρ and as zero on every other irreducible representation. Thus

$$p_\rho p_\sigma = \delta_{\rho\sigma} p_\rho, \qquad \mathbf{1} = \sum_\rho p_\rho.$$

Put $d_\rho = \dim V_\rho$ and $t_\rho = \varepsilon(p_\rho)$. To compute this normalization, let L_z be left multiplication by $z \in \mathbb{C}[G]$ on the regular representation. Every diagonal entry in the group-element basis is $[1_G](z)$, so $\operatorname{tr}(L_z) = |G|[1_G](z)$. The decomposition $\mathbb{C}[G] \cong \bigoplus_\rho V_\rho^{\oplus d_\rho}$ shows that L_{p_ρ} projects onto a subspace of dimension d_ρ^2. Hence

$$t_\rho = \frac{[1_G](p_\rho)}{|G|} = \frac{\operatorname{tr}(L_{p_\rho})}{|G|^2} = \left(\frac{d_\rho}{|G|}\right)^2.$$

The pairing is diagonal in this basis: $\eta(p_\rho, p_\sigma) = \delta_{\rho\sigma} t_\rho$. Its inverse therefore gives

$$\alpha = \sum_\rho t_\rho^{-1} p_\rho^2 = \sum_\rho t_\rho^{-1} p_\rho, \qquad \alpha p_\rho = t_\rho^{-1} p_\rho.$$

By Schur's lemma a class sum C_K acts on V_ρ by a scalar $\lambda_\rho(K)$. Taking traces gives $d_\rho \lambda_\rho(K) = |K|\chi_\rho(K)$, so

$$C_K = \sum_\rho \lambda_\rho(K) p_\rho, \qquad \lambda_\rho(K) = \frac{|K|\chi_\rho(K)}{d_\rho}.$$

When these expansions are multiplied, orthogonality of the idempotents forces all representation indices to agree. Applying the counit then gives

$$\begin{aligned} \omega_{g,n,0}(C_{K_1}, \dots, C_{K_n}) &= \varepsilon\left(\sum_\rho t_\rho^{-g} \prod_{j=1}^n \lambda_\rho(K_j)\, p_\rho\right) \\ &= \sum_{\rho \in \mathrm{Irr}(G)} \left(\frac{d_\rho}{|G|}\right)^{2-2g} \prod_{j=1}^n \frac{|K_j|\chi_\rho(K_j)}{d_\rho}. \end{aligned}$$

Each handle contributes t_ρ^{-1} and the final counit contributes t_ρ, explaining the exponent $1-g$ of t_ρ. For $G = S_d$, take $|G| = d!$ and let K_j be the class of cycle type μ_j; this is exactly the Burnside formula stated earlier.[1]

[1] For the semisimple calculation and the finite-group normalization, see C. Teleman, *Five lectures on Topological Field Theory*, §§1.8–1.10, p. 3.

2 Moduli spaces of curves

Handwritten source, pp. 8–11.

2.1 Smooth pointed curves and stability

Let $g, n \geq 0$ satisfy

$$2 - 2g - n < 0, \qquad \text{equivalently} \qquad 2g - 2 + n > 0.$$

The moduli space $\mathcal{M}_{g,n}$ parametrizes smooth, connected, compact complex curves of genus g with n ordered, pairwise distinct marked points, up to isomorphisms preserving the markings:

$$\mathcal{M}_{g,n} = \{(C; x_1, \ldots, x_n) : C \text{ smooth of genus } g,\ x_i \neq x_j \text{ for } i \neq j\}/\cong .$$

The inequality says that the Euler characteristic of the punctured surface $C \setminus \{x_1, \ldots, x_n\}$ is negative. Here and below, "moduli space" is understood as the moduli stack, or its associated complex orbifold; the displayed quotient describes its isomorphism classes.

The compactification $\overline{\mathcal{M}}_{g,n}$ allows *stable pointed curves*. Such a curve is connected and compact, and satisfies the following conditions:

(i) Its only singularities are nodes, locally of the form $xy = 0$.

(ii) Its marked points are distinct and lie in the smooth locus.

(iii) Every component of its normalization has negative Euler characteristic after its special points are removed. The special points are the markings and the branches of the nodes.

Thus a component of genus h with k special points must satisfy

$$2h - 2 + k > 0.$$

In particular, each rational component has at least three special points, and each genus-one component has at least one. Both branches of a self-node count as special points on the normalization. The genus of a nodal curve here is its *arithmetic genus*, which is the genus of a smooth curve obtained by smoothing all its nodes.

2.2 The example $\mathcal{M}_{0,4}$

A smooth genus-zero curve is isomorphic to $\mathbb{P}^1$. A unique projective coordinate takes the first three marked points to $0, 1, \infty$, so

$$(C; x_1, x_2, x_3, x_4) \cong (\mathbb{P}^1; 0, 1, \infty, t), \qquad \mathcal{M}_{0,4} \cong \mathbb{P}^1 \setminus \{0, 1, \infty\}.$$

In an affine coordinate, the cross-ratio used here is

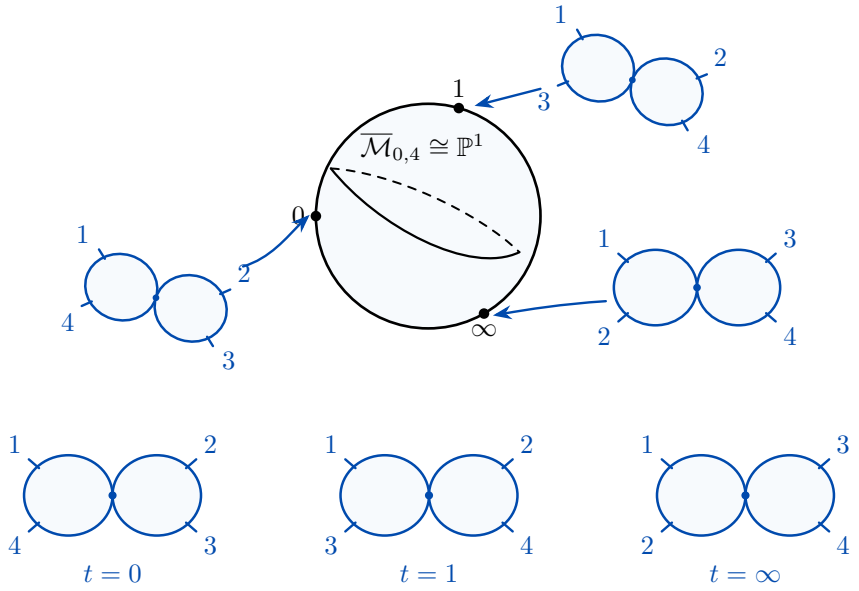
$$t = \frac{x_4 - x_1}{x_2 - x_1} \Big/ \frac{x_4 - x_3}{x_2 - x_3} = \frac{(x_4 - x_1)(x_2 - x_3)}{(x_2 - x_1)(x_4 - x_3)},$$

with the usual limiting interpretation if one of the points is at infinity.

The stable compactification adds three points:

$$\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1.$$

At each added point the stable curve consists of two projective lines meeting in one node, with two markings on each component. Each component then has exactly three special points. The three boundary curves are shown below; a solid dot joining two circles denotes a node.



The corresponding partitions of the labels are $14 \mid 23$, $13 \mid 24$, and $12 \mid 34$, respectively.

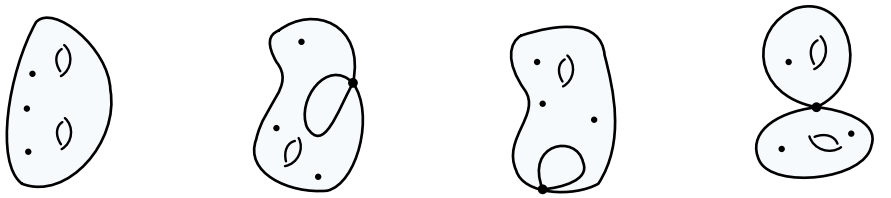
2.3 Degenerations, boundary, and gluing

Allowing nodes records limits of smooth curves. A handle can be pinched to give a nonseparating node, or a separating circle can be pinched to split a curve into two components. The compactified moduli stack is a smooth, proper complex orbifold of complex dimension

$$\dim_{\mathbb{C}} \overline{\mathcal{M}}_{g,n} = 3g - 3 + n.$$

Its boundary $\overline{\mathcal{M}}_{g,n} \setminus \mathcal{M}_{g,n}$ is a normal-crossings divisor. Near a curve with several nodes there is one local smoothing parameter for each node; retaining that node means setting the corresponding parameter equal to zero. Boundary components can also have self-intersections, so the divisor need not have *simple* normal crossings. The smoothness assertion concerns the stack/orbifold, not in general its coarse moduli space.

The following drawings retain the four limiting pictures in the notes. Handles are drawn on the surfaces; a pinched handle gives a nonseparating node, while a separating node divides the curve into two components.



smooth | a nonseparating limit | another nonseparating limit | a separating limit

There are two types of gluing map. For $g = g_1 + g_2$ and $n = n_1 + n_2$,

$$q : \overline{\mathcal{M}}_{g_1,n_1+1} \times \overline{\mathcal{M}}_{g_2,n_2+1} \longrightarrow \overline{\mathcal{M}}_{g,n}$$

identifies the last marked point of the first curve with the last marked point of the second. It produces a separating node.

The map

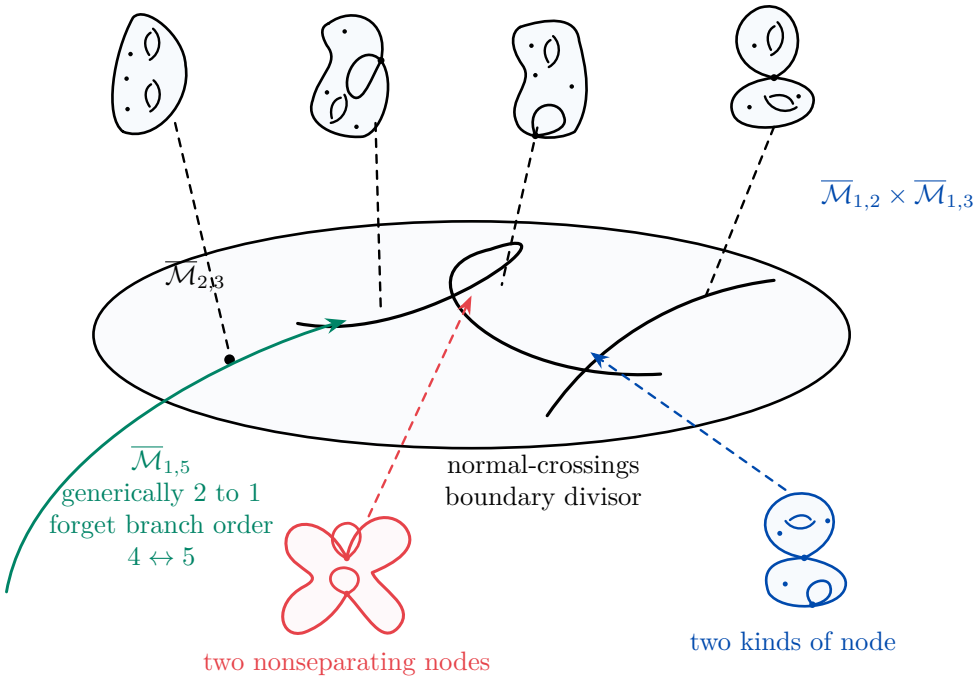
$$s : \overline{\mathcal{M}}_{g-1,n+2} \longrightarrow \overline{\mathcal{M}}_{g,n}$$

identifies the last two marked points of a single connected curve and produces a nonseparating node. All moduli spaces appearing in these maps are required to be stable.

For the example $(g, n) = (2, 3)$ in the handwritten diagram, these include

$$\overline{\mathcal{M}}_{1,2} \times \overline{\mathcal{M}}_{1,3} \longrightarrow \overline{\mathcal{M}}_{2,3}, \qquad \overline{\mathcal{M}}_{1,5} \longrightarrow \overline{\mathcal{M}}_{2,3}.$$

The second map is generically two-to-one onto the nonseparating boundary: swapping markings 4 and 5 gives the same glued curve. It forgets the *ordering of the two branches* at the resulting node.



This is a schematic picture of the boundary, not a list of all its components.

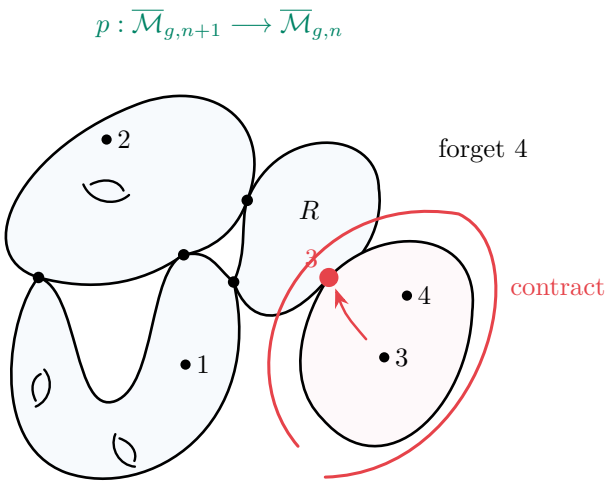
2.4 Forgetting a marking and stabilizing

The forgetful map

$$p : \overline{\mathcal{M}}_{g,n+1} \longrightarrow \overline{\mathcal{M}}_{g,n}$$

forgets the last marking and then contracts any components that have become unstable. For example, a rational tail carrying markings 3 and 4 and attached at one node is stable. After marking 4 is forgotten, it has only two special points, so it is contracted; marking 3 moves to its attaching point on the remaining curve.

In the diagram, the two positive-genus components meet each other at two nodes, and each also meets the rational component R. The component R carries the rational tail. After its contraction, R has two nodes and marking 3, so it remains stable. The incidences of the other components are unchanged.



The red outline indicates the contracted tail; the red 3 marks the new position of the surviving marking.

3 Cohomological field theories

Handwritten source, pp. 11–13.

Let V be a finite-dimensional complex vector space, let $\mathbf{1} \in V$, and let $\eta : V \otimes V \to \mathbb{C}$ be a nondegenerate symmetric bilinear form. Fix a basis (e_μ) and write

$$\eta_{\mu\nu} = \eta(e_\mu, e_\nu), \qquad (\eta^{\mu\nu}) = (\eta_{\mu\nu})^{-1}, \qquad \eta^{-1} = \sum_{\mu,\nu} \eta^{\mu\nu} e_\mu \otimes e_\nu.$$

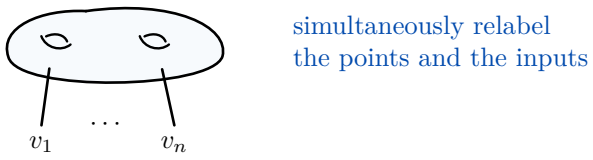
The tensor $\eta^{-1} \in V \otimes V$ is independent of the chosen basis.

Definition 3.1. In the convention of these notes, a *cohomological field theory* (CohFT) on $(V, \eta, \mathbf{1})$ is a collection of linear maps

$$\Omega_{g,n} : V^{\otimes n} \longrightarrow H^{\text{even}}(\overline{\mathcal{M}}_{g,n}; \mathbb{C}), \qquad 2g - 2 + n > 0,$$

satisfying axioms (0), (1), (2a), and (2b) below. The additional flat-unit axiom (3) is recorded separately, as in the handwritten notes.

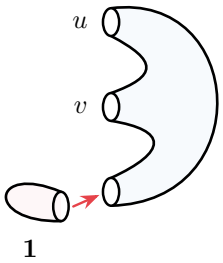
(0) Symmetry. The map $\Omega_{g,n}$ is S_n-equivariant: permuting the vectors agrees with relabeling the marked points. Thus the construction does not privilege one ordering of the markings.



(1) Normalization. Since $\overline{\mathcal{M}}_{0,3}$ is a point, $H^*(\overline{\mathcal{M}}_{0,3}; \mathbb{C}) = \mathbb{C}$, and

$$\Omega_{0,3}(u, v, \mathbf{1}) = \eta(u, v).$$

This expresses the metric using stable moduli spaces and avoids the unstable case $(g, n) = (0, 2)$.



avoid unstable cases

$H^*(\overline{\mathcal{M}}_{0,3}) = \mathbb{C}$
$\overline{\mathcal{M}}_{0,3} = \mathrm{pt}$

(2a) Nonseparating gluing. For $s : \overline{\mathcal{M}}_{g-1,n+2} \to \overline{\mathcal{M}}_{g,n}$,

$$\begin{aligned} s^*\Omega_{g,n}(v_1,\dots,v_n) &= \Omega_{g-1,n+2}(v_1 \otimes \cdots \otimes v_n \otimes \eta^{-1}) \\ &= \sum_{\mu,\nu} \eta^{\mu\nu}\Omega_{g-1,n+2}(v_1,\dots,v_n,e_\mu,e_\nu). \end{aligned}$$

Thus the two branches of the new node are contracted using the inverse metric.



$\eta^{-1} = \sum_{\mu,\nu} \eta^{\mu\nu} e_\mu \otimes e_\nu$

contract the two branches

(2b) Separating gluing. For $q : \overline{\mathcal{M}}_{g_1,n_1+1} \times \overline{\mathcal{M}}_{g_2,n_2+1} \to \overline{\mathcal{M}}_{g,n}$, where $g = g_1 + g_2$ and $n = n_1 + n_2$,

$$\begin{aligned} q^*\Omega_{g,n}(v_1,\dots,v_n) = \sum_{\mu,\nu} \eta^{\mu\nu}\, &\Omega_{g_1,n_1+1}(v_1,\dots,v_{n_1},e_\mu) \\ &\boxtimes \Omega_{g_2,n_2+1}(v_{n_1+1},\dots,v_n,e_\nu). \end{aligned}$$

Here $\boxtimes$ denotes the exterior product of cohomology classes on the two factors. By symmetry the same formula holds for every partition of the marked points.

(3) Flat unit (optional). A CohFT has a *flat unit* if, for the forgetful map $p : \overline{\mathcal{M}}_{g,n+1} \to \overline{\mathcal{M}}_{g,n}$,

$$p^*\Omega_{g,n}(v_1,\dots,v_n) = \Omega_{g,n+1}(v_1,\dots,v_n,\mathbf{1}).$$

The target (g,n) must be stable. The distinction between normalization and this additional condition is retained throughout.

4 The Hodge-bundle example

Handwritten source, pp. 13–15.

4.1 Abelian differentials on nodal curves

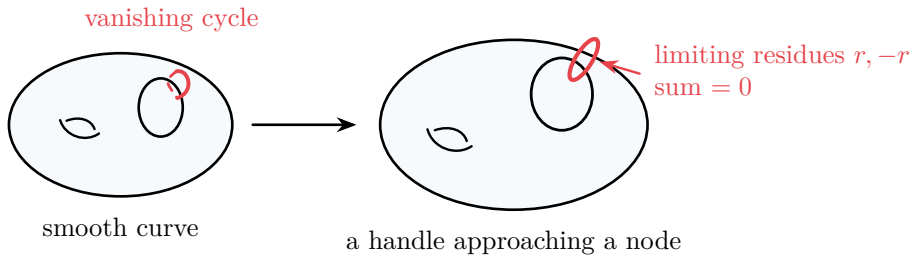
The Hodge bundle

$$\mathbb{E}_{g,n} \longrightarrow \overline{\mathcal{M}}_{g,n}$$

has rank g and fiber

$$\mathbb{E}_C = H^0(C, \omega_C),$$

the space of abelian differentials on the stable curve C. If C is smooth, these are the usual holomorphic 1-forms. On a nodal curve, an abelian differential means a meromorphic differential on each component of the normalization, holomorphic away from the branches of the nodes, with at most simple poles at those branches and with opposite residues at the two branches of each node.



The sum of the residues on every compact normalization component is zero. Individual residues at a nonseparating node need not vanish.

The notation $\mathbb{E}_g$ will suppress n. The statement that the Hodge bundle "does not depend on the markings" means that it is compatible with forgetting and stabilization:

$$p^*\mathbb{E}_{g,n} \cong \mathbb{E}_{g,n+1}$$

whenever both moduli spaces are stable. Contracted rational components contribute no additional abelian differentials.

4.2 The total Chern class defines a CohFT

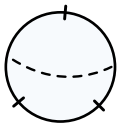
Theorem 4.1. *Take $V = \mathbb{C}$, $\mathbf{1} = 1$, and $\eta(1,1) = 1$. The classes*

$$\Omega_{g,n}(1, \ldots, 1) = c(\mathbb{E}_{g,n}) = 1 + c_1(\mathbb{E}_{g,n}) + \cdots + c_g(\mathbb{E}_{g,n})$$

define a CohFT with flat unit.

Proof. The Hodge bundle is invariant under relabeling the markings, which proves symmetry. In genus zero it has rank zero, so

$$\Omega_{0,3}(1,1,1) = c(\mathbb{E}_{0,3}) = 1 = \eta(1,1).$$



$g = 0: \quad \mathbb{E}_{0,3} = 0$
$c(\mathbb{E}_{0,3}) = 1$

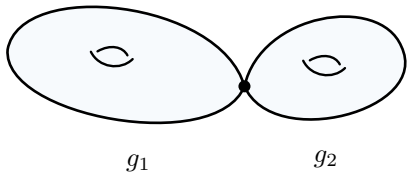
For separating gluing, abelian differentials split into differentials on the two curves being joined. Indeed, at the newly formed separating node the residue on either side is zero: the sum of residues on that connected side is zero, and residues at its internal nodes cancel in pairs. A differential with at most a simple pole and zero residue is regular there. Consequently,

$$q^*\mathbb{E}_{g,n} \cong \mathrm{pr}_1^*\,\mathbb{E}_{g_1,n_1+1} \oplus \mathrm{pr}_2^*\,\mathbb{E}_{g_2,n_2+1}.$$

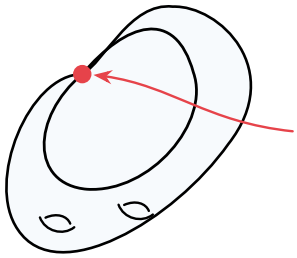
The Whitney sum formula gives

$$q^*c(\mathbb{E}_{g,n}) = c(\mathbb{E}_{g_1,n_1+1}) \boxtimes c(\mathbb{E}_{g_2,n_2+1}),$$

which is axiom (2b).



no pole at the new node
on either connected side



$s^*\mathbb{E}_{g,n}$

residue at the chosen branch a

zero residue $\Longleftrightarrow$
a differential before gluing

For nonseparating gluing, let a, b be the two marked points that are identified. An abelian differential on the glued curve becomes a differential on the original curve with possible simple poles at a, b and residues $r, -r$. Taking the residue at a gives an exact sequence of bundles on $\overline{\mathcal{M}}_{g-1,n+2}$:

$$0 \longrightarrow \mathbb{E}_{g-1,n+2} \longrightarrow s^*\mathbb{E}_{g,n} \xrightarrow{\mathrm{Res}_a} \mathcal{O} \longrightarrow 0.$$

The kernel consists of the differentials regular at both a and b. It has rank $g-1$, whereas the middle bundle has rank g, so the residue map is surjective on each fiber. The last two markings are ordered on the source, which makes the choice of a and thus this trivial quotient well defined. Therefore

$$s^*c(\mathbb{E}_{g,n}) = c(\mathbb{E}_{g-1,n+2})c(\mathcal{O}) = c(\mathbb{E}_{g-1,n+2}),$$

proving axiom (2a). Finally, $p^*\mathbb{E}_{g,n} \cong \mathbb{E}_{g,n+1}$ proves the flat-unit axiom. □

4.3 The degree-zero part

Since $\overline{\mathcal{M}}_{g,n}$ is connected, $H^0(\overline{\mathcal{M}}_{g,n};\mathbb{C}) = \mathbb{C}$. Projecting a CohFT to degree zero gives

$$\omega_{g,n} : V^{\otimes n} \longrightarrow \mathbb{C}, \qquad \omega_{g,n} = [\Omega_{g,n}]_0.$$

The gluing axioms and normalization descend to these scalar-valued maps, giving the stable correlators of a topological field theory. To see the algebra structure, define a product by $\eta(uv, w) = \omega_{0,3}(u, v, w)$. Normalization gives its unit $\mathbf{1}$, and comparing the two separating boundary factorizations of the constant class $\omega_{0,4}$ gives associativity. The Frobenius-algebra construction above then supplies the unstable operations.

For these degree-zero correlators the flat-unit condition is automatic, even if axiom (3) was not imposed on Ω. To see this for $n \geq 1$, attach a rational component carrying v_n and $\mathbf{1}$ to the last marked point of a stable (g,n)-curve. Separating gluing and normalization give

$$\begin{aligned}\omega_{g,n+1}(v_1,\dots,v_n,\mathbf{1}) &= \sum_{\mu,\nu} \eta^{\mu\nu}\omega_{g,n}(v_1,\dots,v_{n-1},e_\mu)\omega_{0,3}(e_\nu,v_n,\mathbf{1}) \\ &= \omega_{g,n}(v_1,\dots,v_n).\end{aligned}$$

There is no dependence on the boundary point because these are constant cohomology classes. The remaining stable case is $n=0$, $g \geq 2$; nonseparating gluing reduces it to the case with markings:

$$\omega_{g,1}(\mathbf{1}) = \sum_{\mu,\nu} \eta^{\mu\nu}\omega_{g-1,3}(e_\mu,e_\nu,\mathbf{1}) = \sum_{\mu,\nu} \eta^{\mu\nu}\omega_{g-1,2}(e_\mu,e_\nu) = \omega_{g,0}.$$

For the Hodge-bundle example, the degree-zero term of every total Chern class is 1, so

$$\omega_{g,n}(1,\dots,1) = 1.$$

$$\Omega_{g,n} : V^{\otimes n} \longrightarrow H^*(\overline{\mathcal{M}}_{g,n})$$

project to degree zero

$$\omega_{g,n} : V^{\otimes n} \longrightarrow H^0(\overline{\mathcal{M}}_{g,n}) = \mathbb{C}$$

Hodge example: $\omega_{g,n}(1,\dots,1) = 1$

4.4 Other multiplicative Hodge classes

Handwritten source, pp. 16.

The same argument applies to

$$\Omega_{g,n}(1,\dots,1) = \exp\left(\sum_{k\geq 1} t_k \operatorname{ch}_k(\mathbb{E}_{g,n})\right)$$

also a CohFT

Indeed, positive Chern characters are additive in the gluing exact sequences, vanish on the trivial line bundle, and are unchanged by forgetting a marking. Exponentiating turns their sums into products. Thus these classes also define rank-one CohFTs. The coefficients t_k are arbitrary; for fixed g,n, only finitely many terms contribute. The index starts at $k=1$: a term involving $\operatorname{ch}_0(\mathbb{E}_{g,n}) = g$ would introduce an extra factor in nonseparating gluing with the metric $\eta(1,1)=1$.

5 Witten's r-spin class

5.1 Roots of the canonical bundle

Handwritten source, pp. 16–17.

Fix an integer $r \geq 2$, a smooth connected projective curve C of genus g, and distinct marked points $x_1, \ldots, x_n$. Write $K_C = \omega_C$ for its canonical (cotangent) line bundle. A root of K_C alone would have degree $(2g-2)/r$, which need not be an integer. The data in the notes are therefore integers

$$\underbrace{a_1, \ldots, a_n \in \{0, \ldots, r-1\}}_{\textit{the label data}}$$

and a line bundle L, together with an isomorphism

$$\begin{array}{c} L \\ \downarrow \\ C \end{array} \qquad \phi : L^{\otimes r} \xrightarrow{\sim} K_C \left(-\sum_{i=1}^{n} a_i x_i \right). \tag{3}$$

Necessarily,

$$\deg L = \frac{2g - 2 - \sum_i a_i}{r} \in \mathbb{Z}. \tag{4}$$

Over $\mathbb{C}$, this degree condition is also sufficient for roots to exist.

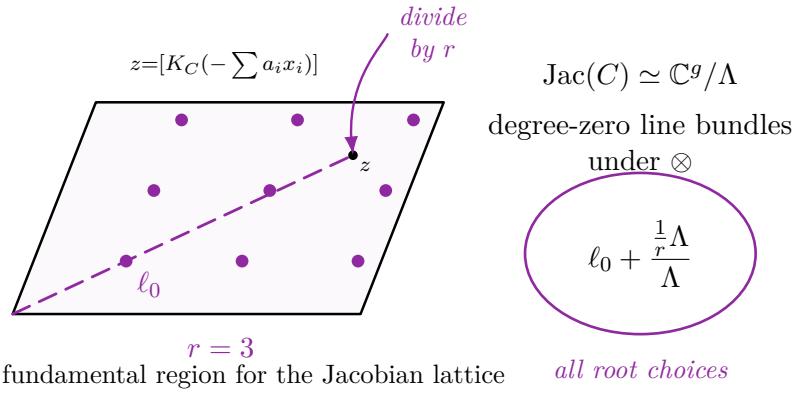
To understand their choices, recall that

$$\mathrm{Jac}(C) = \mathrm{Pic}^0(C) \simeq \mathbb{C}^g/\Lambda,$$

a g-dimensional complex torus; the group operation on degree-zero line bundles is tensor product. First suppose that $K_C(-\sum_i a_i x_i)$ has degree zero, as in the torus picture in the handwritten notes. If its class is represented by $z \in \mathbb{C}^g$, an r-th root is represented by z/r. All choices are

$$\frac{z}{r} + \frac{1}{r}\Lambda \pmod{\Lambda}.$$

In arbitrary degree rd, choose any degree-d line bundle A; then $K_C(-\sum_i a_i x_i) \otimes A^{-r}$ has degree zero, and the same argument applies after tensoring the resulting roots by A.



A fundamental region for the Jacobian lattice (one complex dimension is drawn).

Once one root L_0 is chosen, every other root is $L_0 \otimes T$ for a unique $T \in \mathrm{Jac}(C)[r]$. Thus the root set is a torsor—an affine translate of a group—under

$$\mathrm{Jac}(C)[r] \simeq \frac{\frac{1}{r}\Lambda}{\Lambda} \simeq (\mathbb{Z}/r\mathbb{Z})^{2g}.$$

There are exactly r^{2g} isomorphism classes of roots.

5.2 The moduli of spin curves

Handwritten source, pp. 17–18.

Varying C and its markings gives the moduli of r-spin structures. Over the smooth-curve locus the roots vary without ramification. The count $r^{2g} : 1$ in the notes refers to the cover obtained after removing the scalar automorphisms of a root:

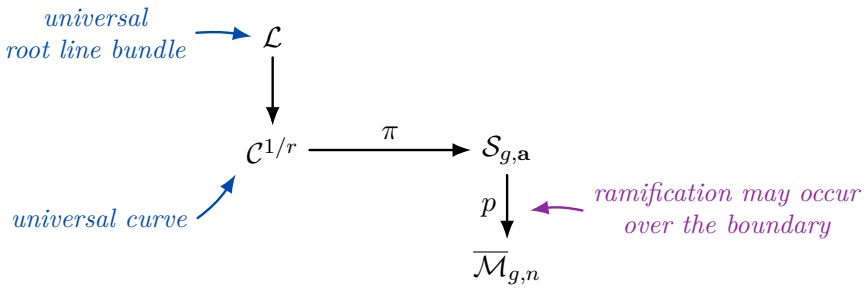


For the actual stack of pairs (L, ϕ), multiplication on L by an r-th root of unity preserves ϕ. Each root therefore has automorphism group μ_r, and the degree of this stack over $\mathcal{M}_{g,n}$ is $r^{2g}/r = r^{2g-1}$.

One compactifies by allowing stable spin curves. The resulting map over $\overline{\mathcal{M}}_{g,n}$ can ramify along the boundary. To retain a universal root *line bundle*, we use the compactification by twisted curves; an equivalent description uses suitable torsion-free roots on nodal curves. Denote the compactified, unrigidified stack by

$$\mathcal{S}_{g,\mathbf{a}} := \overline{\mathcal{M}}^{1/r}_{g;a_1,\dots,a_n}.$$

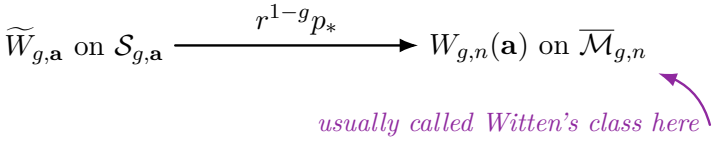
Its universal family and the forgetful map fit into



Here $\mathcal{C}^{1/r}$ is the universal spin curve. Equation (3) describes its fibers over smooth pointed curves, with the corresponding root data understood on twisted fibers at the boundary.

Witten's virtual class is first constructed on $\mathcal{S}_{g,\mathbf{a}}$. The class usually called Witten's class on $\overline{\mathcal{M}}_{g,n}$ is its *normalized* pushforward. Write $\widetilde{W}_{g,\mathbf{a}}$ for the class on the spin stack and set

$$W_{g,n}(a_1, \dots, a_n) := r^{\,1-g} p_* \widetilde{W}_{g,\mathbf{a}}. \tag{5}$$



The factor matches the stack convention just specified; with the rigidified cover of degree r^{2g}, the corresponding factor is r^{-g}.[2] The virtual construction naturally gives a cycle; we use its cohomology class when writing (5).

[2] For the unrigidified normalization and metric, see T. J. Jarvis, T. Kimura and A. Vaintrob, *Moduli Spaces of Higher Spin Curves and Integrable Hierarchies*, Compositio Math. 126 (2001), p. 185, equations (33)–(34), doi:10.1023/A:1017528003622. This factor is implicit in the handwritten pushforward.

5.3 The state space, pairing, and unit

Handwritten source, pp. 18.

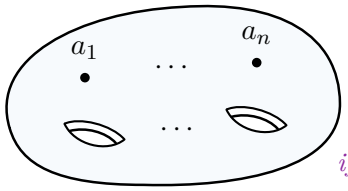
The state space is

$$V = \mathbb{C}\langle e_0, \dots, e_{r-2}\rangle, \qquad \eta(e_a, e_b) = \delta_{a+b,r-2}, \qquad \mathbf{1} = e_0.$$

Thus the matrix of η has 1's on the antidiagonal and 0's elsewhere:

$$\eta = \begin{pmatrix} 0 & \cdots & 0 & 1 \\ \vdots & \cdots & 1 & 0 \\ 0 & 1 & \cdots & \vdots \\ 1 & 0 & \cdots & 0 \end{pmatrix}$$

$\mathbf{1} = e_0$



e_{r-1} is missing

if some $a_i = r-1$, the class vanishes

The formula for η_{ab} also covers $r = 2$, when the matrix is simply (1).

There is no basis vector e_{r-1}. This reflects the vanishing property: if a label $a_i = r - 1$ occurs, Witten's class is zero. For the remaining labels, define

$$\Omega_{g,n}(e_{a_1}, \dots, e_{a_n}) = W_{g,n}(a_1, \dots, a_n).$$

These maps give Witten's r-spin CohFT.

5.4 The simple case: no sections

Handwritten source, pp. 18–19.

Suppose that $H^0(C, L) = 0$ on every fiber of the family in question. Then cohomology and base change, together with Riemann–Roch, imply that

$$\mathcal{V} = R^1\pi_*\mathcal{L}$$

is a vector bundle on the spin moduli space, with fiber $H^1(C, L)$. Indeed, $\chi(C, L) = h^0(C, L) - h^1(C, L)$ is constant, and the hypothesis therefore makes $h^1(C, L)$ constant as well. In this case

$$\widetilde{W}_{g,\mathbf{a}} = c_{\mathrm{top}}(\mathcal{V}^\vee). \tag{6}$$

$\mathcal{V}|_{(C,L)} = H^1(C, L)$ ← *obstruction bundle*

$h^0(C, L) - h^1(C, L) = \chi(C, L)$ *constant by Riemann–Roch*

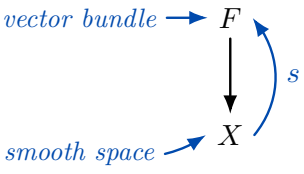
This is the concavity description of Witten's class. The bundle $\mathcal{V}$ is the obstruction bundle for deforming a section of L; the dual in (6) fixes our sign convention.

The geometric idea in the notes is the *fundamental class of the space of sections of L*. The word *virtual* is essential here: the zero section alone need not have the expected codimension, and the obstruction bundle supplies the missing intersection class.

Recall the ordinary model. For a rank-d vector bundle $F \to X$ on a smooth space, a regular section s satisfies

$$c_d(F) \cap [X] = [Z(s)].$$

In the smooth topological setting one uses a transverse section. An algebraic bundle need not have a suitable global section, but its top Chern class still exists.



$$c_{\mathrm{top}}(F) \cap [X]$$
$$= [\text{zero locus of a regular section}]$$

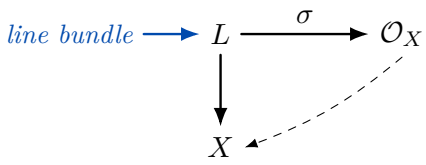
a suitable global section may not exist, but $c_{\mathrm{top}}(F)$ still does

When $H^0(C, L)$ does not vanish, the construction is more difficult. The notes mention Polishchuk–Vaintrob, Chiodo, and the cosection method of Kiem–Li. We next retain the geometric explanation of that method given there.

5.5 Cosections, divisors, and a blow-up

Handwritten source, pp. 19–20.

A *cosection* of a vector bundle is a map from the bundle to the trivial line bundle. In rank one, let



σ: meromorphic cosection
linear on each fiber

be a nonzero meromorphic cosection. This is a meromorphic section of $L^\vee$, linear in each fiber of L. If Z and P are its zero and polar divisors, including multiplicities, then

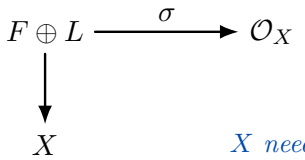
$$c_1(L^\vee) \cap [X] = [Z] - [P].$$

$$c_1(L) \cap [X] \quad = \quad [P] \quad - \quad [Z]$$

Poincaré dual — *poles* — *zeros of the cosection*

The order $P - Z$ is due to the dual: σ is a cosection of L, rather than a section of L. This divisor description works on a noncompact X as well, in its Chow group or in Borel–Moore homology; compactness is not required to define $c_1(L)$.

If σ on $F \oplus L$ vanishes on F, the same picture and the Whitney product formula give



$\sigma|_F = 0$
a cosection only on L

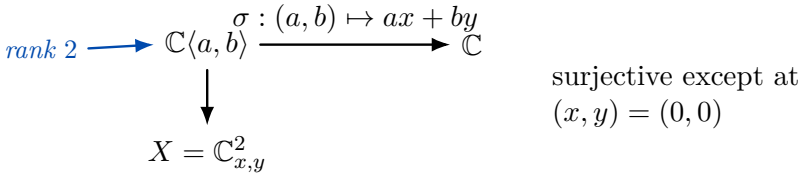
X need not be compact

$$c_{\mathrm{top}}(F \oplus L) \cap [X] = c_{\mathrm{top}}(F) \cap \underbrace{(c_1(L) \cap [X])}_{[P]-[Z]}.$$

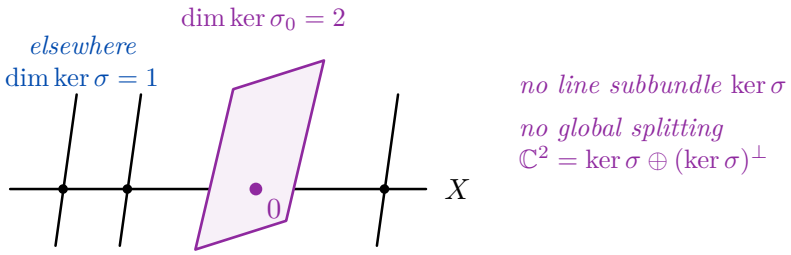
This is the simple model for using a cosection to localize a Chern class, even when X is not compact.

Now take $X = \mathbb{C}^2$, with coordinates (x, y), and the trivial rank-two bundle with fiber coordinates (a, b). Define

$$\sigma : \mathcal{O}_X^{\oplus 2} \longrightarrow \mathcal{O}_X, \qquad \sigma_{(x,y)}(a, b) = ax + by.$$



The map is surjective away from $(0, 0)$, whereas at the origin it is zero. Consequently the fiberwise kernel has dimension 1 away from the origin and dimension 2 at the origin. It is not a line *subbundle* of the original rank-two bundle, so one cannot use a global decomposition into its kernel and a complementary line.



Blow up the origin:

$$b : \widetilde{X} = \mathrm{Bl}_0 \mathbb{C}^2 \longrightarrow \mathbb{C}^2, \qquad E = b^{-1}(0) \simeq \mathbb{P}^1.$$

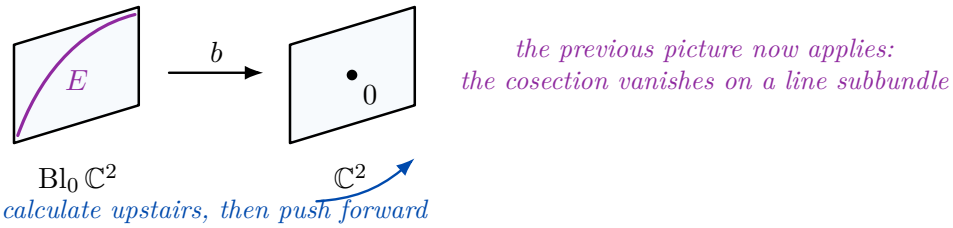
The pulled-back ideal (x, y) is $\mathcal{O}_{\widetilde{X}}(-E)$. Thus the pulled-back cosection factors as

$$\mathcal{O}_{\widetilde{X}}^{\oplus 2} \twoheadrightarrow \mathcal{O}_{\widetilde{X}}(-E) \hookrightarrow \mathcal{O}_{\widetilde{X}},$$

and there is an exact sequence of vector bundles

$$0 \longrightarrow K \longrightarrow \mathcal{O}_{\widetilde{X}}^{\oplus 2} \longrightarrow \mathcal{O}_{\widetilde{X}}(-E) \longrightarrow 0.$$

Here K is now a line subbundle on which the cosection vanishes. The line quotient takes the role of L in the preceding picture; the sequence need not split globally. One can perform the localized intersection calculation upstairs and push it forward by b.



This example explains the role of the blow-up in the notes. For a virtual fundamental class, cosection localization applies this idea to an obstruction theory and its cone; a cosection alone is not a definition of a virtual class.

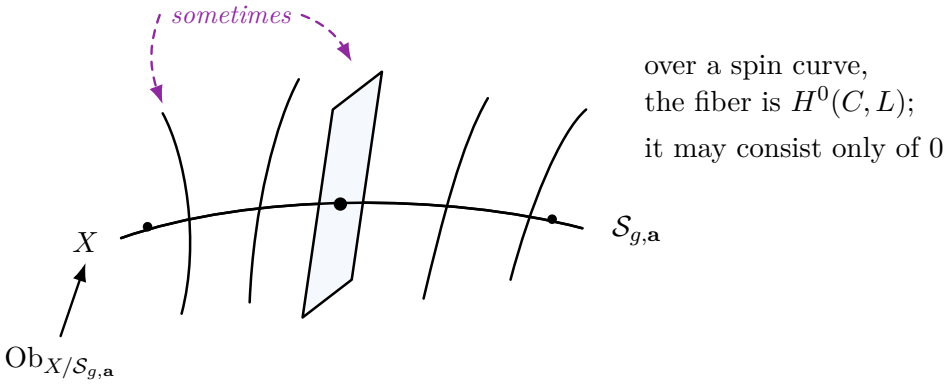
5.6 The cosection associated to a section of the spin bundle

Handwritten source, pp. 21.

Let X denote the moduli of spin curves together with a section:

$$X = \{(C, x_1, \ldots, x_n, L, \phi, s) : s \in H^0(C, L)\}.$$

The zero section identifies the spin moduli space $\mathcal{S}_{g,\mathbf{a}}$ with a closed subspace of X. Where $H^0(C, L) = 0$, the fiber consists only of this zero section. Elsewhere a nonzero space of sections appears, and its dimension can jump.



relative obstruction fiber $H^1(C,L)$

For a section $s \in H^0(C,L)$, equation (3) gives

$$\begin{aligned} s^{r-1} &\in H^0(C, L^{r-1}) \\ &\simeq H^0\left(C, K_C\left(-\sum_i a_i x_i\right) \otimes L^\vee\right) \\ &\subseteq H^0(C, K_C \otimes L^\vee) \end{aligned}$$

forced zero points

The inclusion records the forced zeros at the marked points. Serre duality identifies the last space with

$$H^0(C, K_C \otimes L^\vee) \underset{\text{Serre duality}}{\simeq} H^1(C,L)^\vee.$$

Therefore s^{r-1} defines the fiberwise cosection

$$\sigma_s : H^1(C,L) \longrightarrow \mathbb{C}, \qquad u \longmapsto \langle s^{r-1}, u\rangle_{\mathrm{Serre}}.$$

When the dimensions jump, $H^1(C,L)$ is interpreted through the obstruction sheaf relative to $\mathcal{S}_{g,\mathbf{a}}$, rather than as the fibers of a vector bundle of constant rank. In the spin obstruction theory this relative cosection descends to the obstruction-sheaf cosection used for localization. Its non-surjective locus is the zero-section locus. Cosection localization produces a virtual class supported there, although the full space X of sections need not be proper. This makes precise the virtual interpretation of the space of sections.

We keep the convention (6) throughout. In a convention where the localized obstruction-theoretic class restricts in the concave case to $c_{\mathrm{top}}(\mathcal{V})$, it must be multiplied by $(-1)^D$ to give the class here, since $c_D(\mathcal{V}^\vee) = (-1)^D c_D(\mathcal{V})$. Here $D = -\chi(C,L)$, computed below.[3]

5.7 Degree and the first genus-zero values

Handwritten source, pp. 21–22.

Riemann–Roch and (4) give

$$\begin{aligned} D = -\chi(C,L) &= g - 1 - \frac{2g - 2 - \sum_i a_i}{r} \\ &= \frac{(r-2)(g-1) + \sum_i a_i}{r}. \end{aligned}$$

[3]For the cosection construction and its proper degeneracy locus, see H.-L. Chang, J. Li and W.-P. Li, *Witten's top Chern class via cosection localization*, arXiv:1303.7126, submitted version, pp. 2–3, Definition-Theorem 1.1, and p. 25, Proposition 4.16. That proposition uses the top Chern class of the obstruction bundle itself; the dual convention in the handwritten notes accounts for the sign just stated.

Consequently Witten's class has pure complex degree D:

$$W_{g,n}(a_1,\dots,a_n) \in H^{2D}(\overline{\mathcal{M}}_{g,n};\mathbb{Q}).$$

It is zero if D is not a nonnegative integer. In the concave case, $D = h^1(C,L) = \operatorname{rank}\mathcal{V}$, as required by (6). The notation distinguishes complex degree D from cohomological degree $2D$.

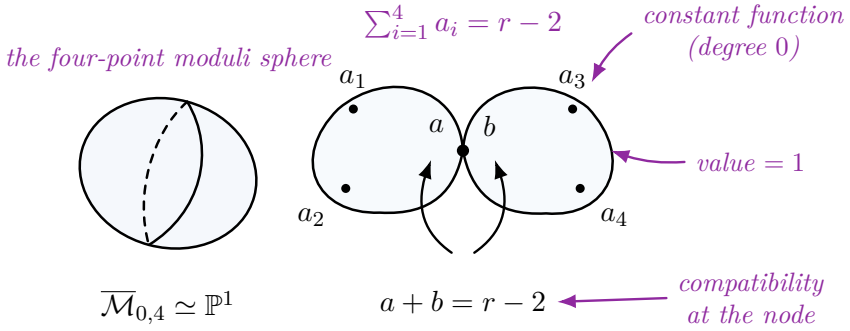
For $a_i \in \{0,\dots,r-2\}$, the three-point value is

$$\boxed{W_{0,3}(a_1,a_2,a_3) = \begin{cases} 1, & a_1+a_2+a_3 = r-2, \\ 0, & \text{otherwise.} \end{cases}}$$

For four points, identify $\overline{\mathcal{M}}_{0,4} \simeq \mathbb{P}^1$, and let $[\mathrm{pt}] \in H^2(\mathbb{P}^1;\mathbb{Q})$ be the class with integral 1. Then the formula in the notes is

$$\boxed{W_{0,4}(a_1,a_2,a_3,a_4) = \begin{cases} 1, & \sum_i a_i = r-2, \\ \dfrac{1}{r}\min\limits_{1\le i\le 4}\{a_i, r-1-a_i\}\,[\mathrm{pt}], & \sum_i a_i = 2r-2, \\ 0, & \text{otherwise.} \end{cases}} \tag{7}$$

The first line is a constant degree-zero class; the second line has cohomological degree 2. Its scalar coefficient is the four-point integral.[4]



The boundary picture explains the constant value in the first line. If $\sum_i a_i = r-2$, gluing on the boundary separating $\{1,2\}$ from $\{3,4\}$ gives

$$\sum_{a+b=r-2} W_{0,3}(a_1,a_2,a)\,W_{0,3}(b,a_3,a_4).$$

There is exactly one nonzero term:

$$a = r-2-a_1-a_2, \qquad b = r-2-a_3-a_4.$$

Both labels lie in $\{0,\dots,r-2\}$, they satisfy $a+b = r-2$, and the two three-point values are 1. The restriction is therefore 1; since the four-point class in this case has degree zero, it is the constant class 1 on $\overline{\mathcal{M}}_{0,4}$.

[4] The three-point integral and the coefficient in the degree-two line are Jarvis–Kimura–Vaintrob, *op. cit.*, Proposition 6.1, pp. 205–206. The factor [pt] in (7) makes explicit that the handwritten scalar represents a degree-two cohomology class.